

Habib Rostami

Publication List–December 27, 2025

The list comprises **52** peer-reviewed publications and **1** preprint currently under review, totaling **53** journal publications together with **8** conference highlights, abstract, comments, and minor erratums and thus in total **61** Research outputs. All papers are accessible via the blue hyperlinks. In addition, my publications are available on [Habib Rostami's Google Scholar](#). The FIVE most cited publications are marked with a red asterisk (*).

Journals: published, arXiv preprint/under review

53. “*Quantum Geometric Injection and Shift Optical Forces Drive Coherent Phonons*”, J. L. Pimlott and **H. Rostami**, [arXiv:2507.17814 \[under review in PRL\]](#).
52. “*Fermi liquid theory of d-wave altermagnets: demon modes and Fano-Demon states*”, **H. Rostami** and J. Hofmann, [Physical Review Letters](#) **135**, 236701 (2025).
51. “*Chiral Phonons*”,
D. M. Juraschek, R. M. Geilhufe, H. Zhu, M. Basini, P. Baum, A. Baydin, S. Chaudhary, M. Fechner, B. Flebus, G. Grissonnanche, A. I. Kirilyuk, M. Lemeshko, S. F. Maehrlein, M. Mignolet, S. Murakami, Q. Niu, U. Nowak, C. P. Romao, **H. Rostami**, T. Satoh, N. A. Spaldin, H. Ueda, and L. Zhang, [Nature Physics](#) **21**, 1532 (2025).
50. “*Hidden order revealed by light-driven Kerr rotation in Centrosymmetric bulk WSe₂*”, E. Cappelluti, **H. Rostami**, and F. Cilento, [npj 2D Materials and Applications](#) **9**, 89 (2025).
49. “*Magnetic-field suppression of tomographic electron transport*”, **H. Rostami**, N. Ben-Shachar, S. Moroz, and J. Hofmann, [Physical Review B](#) **111**, 155434 (2025). [*Editors' Suggestion*]
48. “*Ultrafast dynamics of Rydberg excitons and their optically-induced charged complexes in encapsulated WSe₂ monolayers*”,
A. Genco¹, C. Trovatiello¹, V. Shahnazaryan, O. Dogadov, A. R. Cadore, B. L. T. Rosa, J. A. Kerfoot, T. Ahmed, O. Balci, E. Alexeev, **H. Rostami**, K. Watanabe, T. Taniguchi, S. Tongay, Andrea C. Ferrari, G. Cerullo, and S. Dal Conte, [Nano Letters](#) **25**, 7673 (2025).
47. “*Giant Elasto-Optic Response of Gallium Selenide on Flexible Mica*”,
T. Barker, A. Gray, M. P. Weir, J. S. Sharp, A. Kenton, **H. Rostami**, and A. Patané, [npj Flexible Electronics](#) **9**, 2 (2025).
46. “*Fate of surface gaps in magnetic topological insulators*”,
H. Rostami and Ali G. Moghaddam, [Journal of Physics: Materials](#), **8**, 015001 (2025).

45. "Nonlinear thermoelectric probes of anomalous electron lifetimes in topological Fermi liquids",
J. Hofmann and **H. Rostami**, [Physical Review Research **6** , L042042 \(2024\)](#).
44. "Exciton Spin Hall Effect In Arc-Shaped Strained WSe₂",
A. Shubnic, V. Shahnazaryan, I. A. Shelykh, and **H. Rostami**, [Physical Review B **109**, L201409 \(2024\)](#).
43. "Ultrafast Opto-Electronic and Thermal Tuning of Third-Harmonic Generation in a Graphene Field Effect Transistor",
O. Ghaebi, S. Klimmer, N. Tornow, N. Buijssen, T. Taniguchi, K. Watanabe, A. Tomadin, **H. Rostami**, and G. Soavi, [Advanced Science **11**, 2401840 \(2024\)](#).
42. "Theory of pump-driven opto-magnetic properties in semiconducting transition-metal dichalcogenides: an analytical model",
H. Rostami, F. Cilento, and E. Cappelluti, [Nanomaterials **14**, 707 \(2024\)](#).
41. "Light-induced nonlinear spin Hall current in single-layer WTe₂",
P. Bhalla and **H. Rostami**, [New Journal of Physics **26**, 023042 \(2024\)](#).
40. "Flat-band optical phonons in twisted bilayer graphene",
E. Cappelluti, J. A. Silva-Guillén, and **H. Rostami**, and F Guinea, [Physical Review B **108**, 125401 \(2023\)](#).
39. "Light-induced shear phonon splitting and instability in bilayer graphene",
H. Rostami, [Physical Review B **107**, 165418 \(2023\)](#).
38. "Theory for shear displacement by light-induced Raman force in bilayer graphene",
H. Rostami, [Physical Review B **106**, 155405 \(2022\)](#).
37. "Second harmonic helicity and Faraday rotation in gated single-layer 1T'-WTe₂",
P. Bhalla and **H. Rostami**, [Physical Review B **105**, 235132 \(2022\)](#).
36. "Strain-driven chiral phonons in two-dimensional hexagonal materials",
H. Rostami, F. Guinea, E. Cappelluti, [Physical Review B **105**, 195431 \(2022\)](#).
35. "Pseudogauge field driven acoustoelectric current in two-dimensional hexagonal Dirac materials",
P. Bhalla, G. Vignale, and **H. Rostami**, [Physical Review B **105**, 125407 \(2022\)](#);
[Erratum Phys. Rev. B **106**, 079904 \(2022\)](#).
34. "Probing Electronic States in Monolayer Semiconductors through Static and Transient Third-Harmonic Spectroscopy",
Y. Wang, F. Iyikanat, **H. Rostami**, X. Bai, X. Hu, S. Das, Y. Dai, L. Du, Y. Zhang, S. Li, H. Lipsanen, F. J. G. de Abajo, and Z. Sun [Advanced Materials **34**, 2107104, \(2021\)](#).
33. "Nonlinear exciton drift in piezoelectric two-dimensional materials",
V. Shahnazaryan and **H. Rostami**, [Physical Review B **104**, 085405 \(2021\)](#).
32. "Gauge invariance and Ward identities in nonlinear response theory",
H. Rostami, M. I. Katsnelson, G. Vignale, and M. Polini , [Annals of Physics **431**, 168523 \(2021\)](#).

31. “Many-Body Effects in Third Harmonic Generation of Graphene”,
H. Rostami and E. Cappelluti, [Physical Review B **103**, 125415 \(2021\)](#).
30. “Dominant role of two-photon vertex in nonlinear response in two-dimensional Dirac systems”,
H. Rostami and E. Cappelluti, [npj 2D Materials and Applications **5**, 50 \(2021\)](#).
29. “Acoustogalvanic effect in Dirac and Weyl semimetals”,
P. O. Sukhachov and **H. Rostami**, [Physical Review Letters **124**, 126602 \(2020\)](#).
28. “Probing quantum criticality using nonlinear Hall effect in a metallic Dirac system”,
H. Rostami and V. Juričić, [Physical Review Research **2**, 013069 \(2020\)](#).
27. “Layer and orbital interference effects in photoemission from transition metal dichalcogenides”,
H. Rostami, K. Volckaert, N. Lanata, S. K. Mahatha, C. E. Sanders, M. Bianchi, D. Lizzit, L. Bignardi, S. Lizzit, J. A. Miwa, A. V. Balatsky, P. Hofmann, and S. Ulstrup, [Physical Review B **100**, 235423 \(2019\)](#).
26. “Momentum-resolved linear dichroism in bilayer MoS₂”,
K. Volckaert*, **H. Rostami***, D. Biswas, I. Marković, F. Andreatta, C. E. Sanders, P. Majchrzak, C. Cacho, R. T. Chapman, A. Wyatt, E. Springate, D. Lizzit, L. Bignardi, S. Lizzit, S. K. Mahatha, M. Bianchi, N. Lanata, P. D. C. King, J. A. Miwa, A. V. Balatsky, P. Hofmann, and S. Ulstrup; *Equally contributed., [Physical Review B **100**, 241406\(R\) \(2019\)](#).
25. “Hot electrons modulation of third harmonic generation in graphene”,
G. Soavi, G. Wang, **H. Rostami**, A. Tomadin, O. Balci, I. Paradeisanos, E.A.A. Pogna, G. Cerullo, E. Lidorikis, M. Polini, and A. C. Ferrari, [ACS Photonics **6**, 11, 2841 \(2019\)](#).
24. “Exciton routing in the heterostructure of a transition metal dichalcogenide monolayer on a paraelectric substrate”
V. Shahnazaryan, O. Kyriienko, and **H. Rostami**, [Physical Review B **100**, 165303 \(2019\)](#).
23. “Transient hot electron dynamics in single-layer TaS₂”,
F. Andreatta*, **H. Rostami***, A. G. Čabo, M. Bianchi, C. E. Sanders, D. Biswas, C. Cacho, A. J. H. Jones, R. T. Chapman, E. Springate, P. D. C. King, J. A. Miwa, A. V. Balatsky, S. Ulstrup, and P. Hofmann; *Equally contributed. , [Physical Review B **99**, 165421 \(2019\)](#).
22. “Pseudo-Euler equations from nonlinear optics: plasmon-assisted photodetection beyond hydrodynamics”,
A. Principi, D. Bandurin, **H. Rostami**, and M. Polini, [Physical Review B **99**, 075410 \(2019\)](#).
21. “Helical metals and insulators: Sheet singularity of the inflated Berry monopole”,
H. Rostami, E. Cappelluti, and A.V. Balatsky, [Physical Review B **98**, 245114 \(2018\)](#).
20. “Nonlinear anomalous photocurrents in Weyl semimetal”,
H. Rostami and M. Polini , [Physical Review B **97**, 195151 \(2018\)](#).

19. * “Broadband, electrically tuneable, third harmonic generation in graphene”,
G. Soavi, G. Wang, **H. Rostami**, D. Purdie, D. De Fazio, T. Ma, B. Luo, J. Wang, A.K. Ott, D. Yoon, S. Bourelle, J.E. Muench, I. Goykhman, S. Dal Conte, M. Celebrano, A. Tomadin, M. Polini, G. Cerullo, and A.C. Ferrari, [Nature Nanotechnology](#) **13**, 583 (2018).
18. “Piezoelectricity and valley Chern number in inhomogeneous hexagonal 2D crystals”,
H. Rostami, F. Guinea, M. Polini and R. Roldán, [npj 2D Materials and Applications](#) **2**, 15 (2018).
17. * “Ultra-strong nonlinear optical processes and trigonal warping in MoS₂ layers”,
A. Saynatjoki*, L. Karvonen*, **H. Rostami***, A. Autere, S. Mehravar, A. Lombardo, R. A. Norwood, T. Hasan, N. Peyghambarian, H. Lipsanen, K. Kieu, A. C. Ferrari, M. Polini, and Z. Sun; *Equally contributed, [Nature Communications](#) **8**, 893 (2017).
16. “Impurity effects and bandgap-closing in massive Dirac systems”,
H. Rostami and E. Cappelluti, [Physical Review B](#) **96**, 054205 (2017).
15. “Theory of plasmonic effects in nonlinear optics: the case of graphene”,
H. Rostami, M. I. Katsnelson and M. Polini, [Physical Review B](#) **95**, 035416 (2017).
14. “Spin relaxation and the Kondo effect in transition metal dichalcogenide monolayers”,
H. Rostami, A. G. Moghaddam, and R. Asgari, [Journal of Physics:Condensed Matter](#) **28**, 505002 (2016).
13. “Edge modes in zigzag and armchair ribbons of monolayer MoS₂”,
H. Rostami, R. Asgari, and F. Guinea, [Journal of Physics:Condensed Matter](#) **28**, 495001 (2016).
12. “Theory of third-harmonic generation in graphene: A diagrammatic approach”,
H. Rostami and M. Polini, [Physical Review B](#) **93**, 161411(R) (2016).
11. * “Theory of strain in single-layer transition metal dichalcogenides”,
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10. “Charge compressibility and quantum magnetic phase transition in MoS₂”,
H. Rostami and R. Asgari, [Physical Review B](#) **91**, 235301 (2015).
9. * “Valley Zeeman effect and spin-valley polarized conductance of monolayer MoS₂ in perpendicular magnetic field”,
H. Rostami and R. Asgari, [Physical Review B](#) **91**, 075433 (2015).
8. “Intrinsic optical conductivity of modified-Dirac fermions system”,
H. Rostami and R. Asgari , [Physical Review B](#) **89**, 115413 (2014).
7. “Andreev reflection in monolayer MoS₂”,
L. Majidi, **H. Rostami**, and R. Asgari, [Physical Review B](#) **89**, 045413 (2014).
6. “Valley polarized transport in strained graphene based Corbino disc”,
Z. Khatibi, **H. Rostami**, and R. Asgari, [Physical Review B](#) **88**, 195426 (2013).

5. “Electronic structure and layer-resolved transmission of strained bilayer graphene in the presence of vertical fields”,
H. Rostami and R. Asgari, [Physical Review B **88**, 035404 \(2013\)](#).
4. * “Effective lattice Hamiltonian for monolayer MoS₂: Tailoring electronic structure with perpendicular electric and magnetic fields”,
H. Rostami, A. G. Moghaddam, and R. Asgari, [Physical Review B **88**, 085440 \(2013\)](#).
3. “Indirect exchange interaction between magnetic adatoms in monolayer MoS₂”,
F. Parhizgar, **H. Rostami**, and R. Asgari, [Physical Review B **87**, 125401 \(2013\)](#).
2. “Electronic ground state properties in strained graphene”,
H. Rostami and R. Asgari, [Physical Review B **86**, 155435 \(2012\)](#).
1. “Fabrication of optical magnetic mirrors using bent and mushroom-like carbon nanotubes”,
H. Rostami, Y. Abdi, and E. Arzi, [Carbon **48**, 3659 \(2010\)](#).

Conferences, Comments, Highlights

8. “Erratum: Pseudogauge field driven acoustoelectric current in two-dimensional hexagonal Dirac materials [Phys. Rev. B (2022) 105 (125407)]”, [Physical Review B **106**, 079904, \(2022\)](#)
7. “Highlights from Faraday Discussion: From optical to THz control of materials, London, UK, May 2022”,
A.V. Girija, M. Basini, and **H. Rostami**, [Chemical Communications **59**, 3939 \(2023\)](#).
6. “THz and laser field excitation processes: general discussion”,
M. Basini, L. Benfatto, E. Collet, A. V Girija, G. Huitric, S. Iwai, J. O. Johansson, S. L. Johnson, S. Koshihara, R. V. Mikhaylovskiy, J. M. Rost, **H. Rostami**, and E. Sutcliffe, [Faraday Discussions **237**, 406 \(2022\)](#).
5. “Theory of out of equilibrium light-induced phenomena: general discussion”,
M. Alías-Rodríguez, M. Basini, L. Benfatto, Y. Boeije, I. Burghardt, A. Burnett, L. Chen, E. Collet, R. Cowin, I. Eremin, G. Fleming, A. V Girija, K. Ishida, S. Iwai, J. O. Johansson, S. L. Johnson, K. Katsumi, J. McCusker, C. Odin, M. Puviani, J. M. Rost, **H. Rostami**, M. Udina, and J. Weinstein, [Faraday Discussions **237**, 198 \(2022\)](#).
4. “Material science: ultrafast transformation, electron–phonon coupling, multi-scale aspects: general discussion”,
M. Basini, L. Benfatto, Y. Boeije, W. Bronsch, L. X. Chen, E. Collet, A. V. Girija, H. Hedayat, K. Ishizaka, S. Iwai, J. O. Johansson, S. L. Johnson, K. Katsumi, S. Koshihara, J. M. Rost, **H. Rostami**, J. Stähler, M. Udina, [Faraday Discussions **237**, 80 \(2022\)](#).
3. “Many-Body Theory of Nonlinear Response in Dirac Materials”,
H. Rostami and E. Cappelluti, [APS March Meeting 2021, abstract id.J41.001 \(2021\)](#)
2. “Graphene electrically tuneable third harmonic generation”,
G. Soavi, G. Wang, **H. Rostami**, D. Purdie, D. De Fazio, T. Ma, B. Luo, J. Wang, A.K.

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1. “*Phases of Interacting Dirac Matter*”,
H. Rostami, S. Banerjee, G. Aeppli, A. Balatsky, [APS March Meeting Abstracts V36.013](#) (2018)